

AHMED RAYHAN

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[LinkedIn](#) | [ResearchGate](#) | [Web](#) | [Google Scholar](#)

Education

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Science in Electrical and Electronic Engineering
CGPA 3.559/4.00

Dhaka, Bangladesh
January 2025

HIGHER SECONDARY SCHOOL CERTIFICATE

Birshrestha Noor Mohammad Public College
GPA 5.0/5.0

Dhaka, Bangladesh
July 2019

SECONDARY SCHOOL CERTIFICATE

Birshrestha Noor Mohammad Public College
GPA 5.0/5.0

Dhaka, Bangladesh
June 2017

Research & Publication

- M. Sabbir Hasan Sohag, **A. Rayhan**, et al., “Numerical Analysis of a LiF-Ge based 1D Photonic Crystal Biosensor for Detection of Cancer Cells,” Journal of Optics, vol. 27, p. 115001, Nov 2025. [\[Link\]](#)
- **A. Rayhan** and Z. I. Bushra, “Enhanced Blood Glucose Detection Using SPR-based Biosensor with CdS/Ag/CdS/BP Nanomaterials,” Physica Scripta, vol. 100, p. 075517, Jun 2025. DOI: 10.1088/1402-4896/addfb3. [\[Link\]](#)
- **A. Rayhan**, S. A. Rafi, N. A. Zaman, and W. Emon, “Detection of Tuberculosis in Blood Samples Using One-dimensional Photonic Crystal,” Optical and Quantum Electronics, vol. 57, no. 8, p. 436, 2025. DOI: <https://doi.org/10.1007/s11082-025-08347-1>. [\[Link\]](#)
- Juwel, M., Al Amin, M., **A. Rayhan** et al. “Niobium Pentoxide and Black Phosphorus Based Multilayer Surface Plasmon Resonance Biosensor for the Detection of Ultra-sensitive Mycobacterium Tuberculosis.” *J Comput Electron* 25, 55 (2026). <https://doi.org/10.1007/s10825-025-02489-x> [\[Link\]](#)
- **A. Rayhan**, et al., "Performance Analysis of Free Space Optical Communication in Urban Rain Environment," 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2025, pp. 1-6, doi: 10.1109/ECCE64574.2025.11013976. [\[Link\]](#)

Experience

Bangladesh University of Business and Technology

Department of Electrical and Electronic Engineering

Lecturer

Dhaka, Bangladesh

July 2025- Present

- Conducting undergraduate courses. Responsibilities include delivering lectures, grading assignments, evaluating students' research, developing course materials, conducting research, and contributing to administrative tasks and departmental committees.

Projects

- DC Motor **Speed and Direction** Control using Mosfets
- Implemented **Schematic and Layout** using virtuoso to implement Conventional Fat tree encoder.
- Home Security System using FSM and executed **Full Chip Design** with cadence Innovus & Genus.
- Automatic **Voltage Stabilizer**: Built device to regulate AC power output.
- LED Thermometer: Constructed a digital thermometer using **analog ICs**.
- **RFID** Locking System: Designed access control system.
- Gas Leakage Detection: Created **IoT** based system for gas leak detection.

Internships

Neural Semiconductor Ltd.

Dhaka, Bangladesh

Attachment Internship

July 2024

- Gained a comprehensive overview of the ASIC design flow and industrial chip fabrication through professional-led workshops that bridged academic theory with real-world application.

Dhaka Power Distribution Company Ltd.

Dhaka, Bangladesh

Attachment Internship

October 2023 – November 2023

- Gained practical experience in power distribution and grid operations at DPDC, focusing on SCADA systems, voltage switching, and the industrial application of AIS/GIS systems.

Leadership and Activities

AUST Career Development Club

Dhaka, Bangladesh

Executive Director

April 2024 – December 2024

- Manage all IT department functions: strategic planning, project management, and team supervision.
- Lead major event planning and execution, maximizing attendee engagement.

Huawei Seeds For the Future

April 2024-July 2024

- Developed leadership, soft skills, and a winning IoT-based SDG project while networking with industry professionals.

Test Scores

IELTS Academic: Overall 7.5

Test Date: November 1, 2025

Honors & Awards

- Achieved top 5% academic ranking in Spring '20, Fall '20, and Fall '23 semesters.
- Champion of IEEE's **Research Revelry: EV batteries**
Issued by IEEE AUST STUDENT BRANCH · May 2024
- Champion of IEEE's **Poster Pavilion – A Showcase of Innovation**
Issued by IEEE AUST STUDENT BRANCH · Jun 2023
- 8th Place in **Research Article Writing Competition**
Issued by IEEE AUST STUDENT BRANCH · Aug 2023

Skills & Interests

Technical	: MATLAB, COMSOL Multiphysics, CST Studio, Machine Learning, Python, C++, Virtuoso, Innovus, Genus, Arduino.
Soft-skills	: Problem Solving, Time Management, Leadership, Adaptability.
Interests	: Researching cutting-edge technology, reading fictions & travelling.

References

Dr. Bobby Barua
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Department of Electrical and Electronic Engineering
Ahsanullah University of Science and Technology
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Dr. Ahmed Al Mansur
Chairman and Professor,
Department of Electrical and Electronic Engineering,
Bangladesh University of Business and Technology
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